

Table 2: Optimization Dataset Statistics

Group	$N_{playlist}$	H_{avg}	N_{Track}	N_{Artist}
K=0	1000	38	0	0
K=1	1000	37	942	758
K=5	2000	33	7548	3496
K=10	2000	33	13487	5127
K=25	2000	32	27185	8789
K=100	2000	53	76648	18242

980001 – 990000 from the MPD). Similarly to the original challenge dataset, we divided these playlists into 10 distinct categories that match the challenge categories (see Section 2) via random sampling. The statistics of the D_{opt} dataset can be seen in Table 2.

The optimization process was set maximize the NDCG metric (Equation 3) and was executed using Tree-structured Parzen Estimator (TPE) [4], which is a type of a Sequential Model-Based Global Optimization (SMBO) [7] algorithm. We use the TPE implementation that is available in the Python library Hyperopt [3]. The TPE optimization process was set to run for 100 iterations and the search space of weights defined as a uniform random variable ranging from 0 to 1.

Table 3 shows the optimized best sets of weights separately for each category (used in the local weights composition) and global weights (used in the global weights composition).

Table 3: Optimized weights

Category	W_u	W_w	W_{al}	W_{ar}	m
title_only	0.000	1.000	0.000	0.000	-
1_with_title	1.000	0.423	0.001	0.011	1
5_no_title	1.000	0.000	0.040	0.040	2
5_with_title	1.000	0.337	0.006	0.010	2
10_no_title	1.000	0.000	0.003	0.009	2
10_with_title	1.000	0.964	0.002	0.001	2
25_first	1.000	0.795	0.001	0.037	2
25_random	1.000	0.437	0.145	0.022	2
100_first	1.000	0.828	0.028	0.045	2
100_random	0.915	1.000	0.169	0.133	3
global	1.000	0.517	0.084	0.056	-

Model Selection

In order to select the best model from the proposed framework in an offline manner (without making an official submission), we extracted a validation dataset (D_{val}) containing 10k playlists (playlists 990001 – 1000000 from the MPD). Again, we divided these playlists into ten distinct challenge categories via random sampling. The statistics of the D_{val} dataset

Table 4: Validation Dataset Statistics

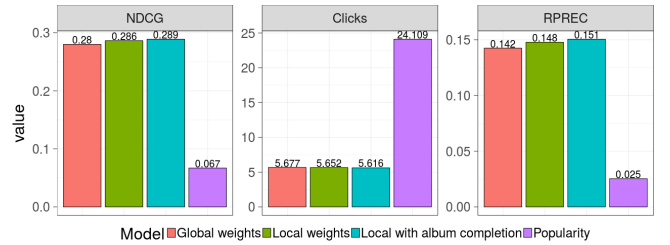
Group	$N_{playlist}$	H_{avg}	N_{Track}	N_{Artist}
K=0	1000	38	0	0
K=1	1000	38	943	737
K=5	2000	34	7451	3427
K=10	2000	33	13455	5197
K=25	2000	33	26864	8512
K=100	2000	53	80171	18570

can be seen in Table 4. The validation set was used as a proxy to the challenge leaderboard, guiding model selection and improvements.

Results

This subsection presents and discusses the results of our experiments.

Figure 1 shows the performance (in terms of NDCG, CLICKS, and RPREC) for the tested instantiations of the framework and the baseline popularity model. Note that in all cases, the composed collaborative model performs better than the popularity model. The model with local weights and album completion is the best performing model and was the selected strategy for our final submission.

**Figure 1: Performance of different models on validation set**

The results of the final model on both the validation set (D_{val}) and the challenge set are presented in Table 5. The Leaderboard score is the score given by the submission website, calculated by the organizers based on the recommended tracks and the holdout tracks (the ground truth values not available to participants) in the challenge dataset.

Table 5: Results of Composed Model

Metric	Validation	Leaderboard
RPREC	0.150587	0.203652
NDCG	0.288921	0.361175
CLICKS	5.6156	2.0240

To further analyse the performance of the final model within different challenge categories, Table 6 presents the results for the composed model in each of these categories in the validation set¹. We can see in this table that the model is doing considerably better in the groups where the seed tracks were selected randomly from the playlist. The performance is lowest in the category where only the playlist title was provided as input.

Table 6: Results by challenge category (model trained on 400k playlists, 100k tracks)

Category	NDCG	CLICKS	RPREC
title_only	0.179	13.990	0.092
1_with_title	0.266	7.252	0.141
5_no_title	0.270	6.373	0.135
5_with_title	0.271	5.943	0.135
10_no_title	0.259	6.740	0.128
10_with_title	0.263	6.565	0.130
25_first	0.258	6.353	0.122
25_random	0.353	3.307	0.199
100_first	0.219	6.093	0.108
100_random	0.372	2.782	0.223
Overall	0.271	6.540	0.141

5 CONCLUSION

In the 2018 RecSys challenge, teams competed in the task of automatic playlist competition. To simulate different challenges in the playlist completion task, a challenge dataset was provided with ten different types of seed information (called challenge categories). Our solution was based on combining multiple different collaborative filters that each capture different aspects of a playlist, and we combined them using a Tree-structured Parzen Estimator optimization approach where we optimized the weights locally for each of the challenge categories. The solution strategy shows promising results, ranking our team in position 12 out of 112 teams in the final competition leaderboard.

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¹Note that the overall scores are slightly different than in the above, since this detailed evaluation was executed with training on 400k playlists and a total of 100k tracks only, to reduce the computations.